INCH-POUND

A–A–59590/6A
28 April 2005
SUPERSEDING
A-A-59590/6
17 September 2001

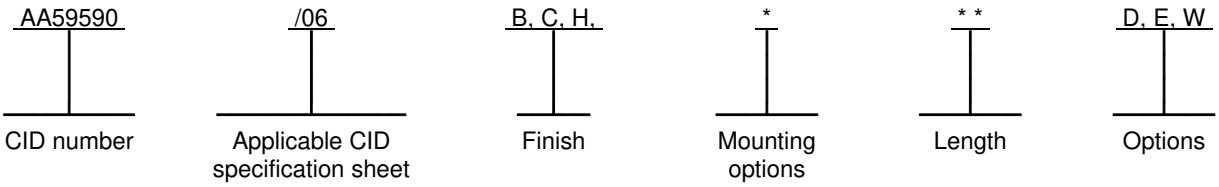
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE 225 X .260 INCH BODY SIZE, WITH VISUAL LOCK AND SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes).



SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces. Card holders shall have a visual lock indicator to show when the assembly is in its relaxed (unlocked) state. The screw retention feature is to prevent unintentional disassembly of the card holder.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, 5, in table 1, and A–A–59590.

Finish options. The finish options “B” (black anodize), “C” (chemical film), or “H” (hard black anodize) shall be as specified in A–A–59590.

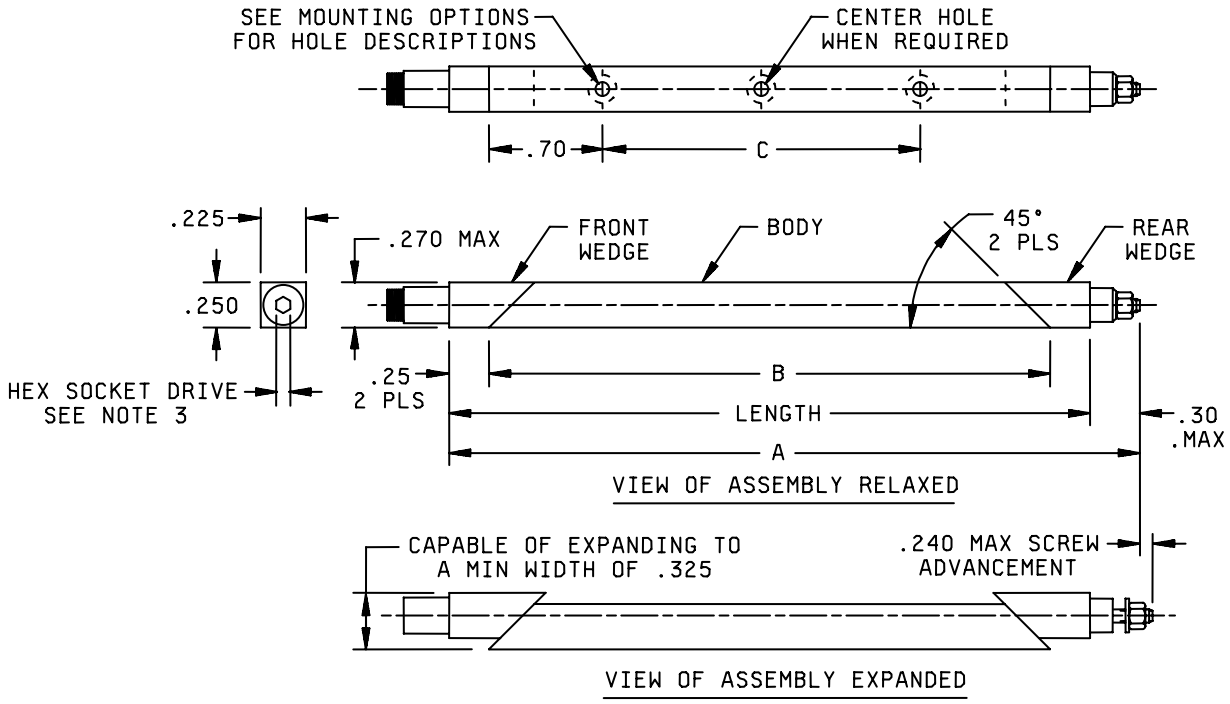
Mounting options. The mounting options shall be as specified in A–A–59590. The options available are as follows: “J” or “K” (rivet mounting holes), “N” (no mounting holes), “T” or “R” (tapped 0-80 UNF 2B holes), “V” or “S” (tapped 2-56 UNC 2B holes), or “G” or “M” (tapped M2.5 x.45 metric holes). See table I and figure 1 for mounting hole spacing requirements. Card holders using mounting option “J” or “K” may be shipped unassembled.

Mounting holes (when required). Two mounting holes are required on card holders less than 6.00 inches (152.4 mm) in length. Three mounting holes are required on card holders 6.00 inches (152.4 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See table I and figure 1 for mounting hole locations.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

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Length, expanded and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths are listed in table I.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.225	5.72	.250	6.35	.270	6.86	.325	8.26
.240	6.09	.25	6.35	.30	7.6	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.51 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.

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Screw retention (locknut). The screw retention feature intended to prevent unintentional disassembly of the card holder shall be as specified in figure 2. The locknut shall be fabricated of a corrosion resistant material.



FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

Visual lock indicator (see figure 3). When the card holder is in the unlocked (relaxed) state, the actuating end of the screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

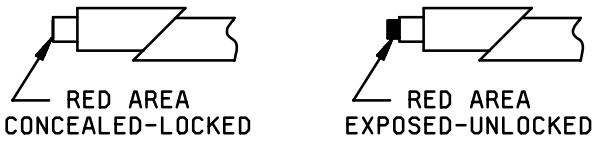


FIGURE 3. Visual lock indicator.

TABLE I. Additional assembly dimensions (see figures 1 and 2). 1/

PIN length designator	Dimension "LENGTH" ± .02 (0.5)	Dimension "A" +.02 / –.09 (+0.5 / –2.3)		Dimension "B" ± .03 (0.8)	Dimension "C" ± .005 (0.13)
		Without options W or D	With options W or D		
18	1.8 (45.7)	2.30 (58.4)	2.40 (60.9)	1.30 (33.0)	.75 (19.1)
23	2.3 (58.4)	2.80 (71.1)	2.90 (73.7)	1.80 (45.7)	.40 (10.2)
28	2.8 (71.1)	3.30 (83.8)	3.40 (86.4)	2.30 (58.4)	.90 (22.9)
30	3.0 (76.2)	3.50 (88.9)	3.60 (91.4)	2.50 (63.5)	1.10 (27.9)
33	3.3 (83.8)	3.80 (96.5)	3.90 (99.1)	2.80 (71.1)	1.40 (35.6)
35	3.5 (88.9)	4.00 (101.6)	4.10 (104.1)	3.00 (76.2)	1.60 (40.6)
38	3.8 (96.5)	4.30 (109.2)	4.40 (111.8)	3.30 (83.8)	1.90 (48.3)
43	4.3 (109.2)	4.80 (121.9)	4.90 (124.5)	3.80 (96.5)	2.40 (61.0)
48	4.8 (121.9)	5.30 (134.6)	5.40 (137.2)	4.30 (109.2)	2.90 (73.7)
53	5.3 (134.6)	5.80 (147.3)	5.90 (149.9)	4.80 (121.9)	3.40 (86.4)
55	5.5 (139.7)	6.00 (152.4)	6.10 (154.9)	5.00 (127.0)	3.60 (91.4)
58	5.8 (147.3)	6.30 (160.0)	6.40 (162.6)	5.30 (134.6)	3.90 (99.1)
61	6.1 (154.9)	6.60 (167.6)	6.70 (170.2)	5.60 (142.2)	4.20 (106.7)
63	6.3 (160.0)	6.80 (172.7)	6.90 (175.3)	5.80 (147.3)	4.40 (111.8)
67	6.7 (170.2)	7.20 (182.9)	7.30 (185.4)	6.20 (157.5)	4.80 (121.9)
68	6.8 (172.7)	7.30 (185.4)	7.40 (188.0)	6.30 (160.0)	4.90 (124.5)
73	7.3 (185.4)	7.80 (198.1)	7.90 (200.7)	6.80 (172.7)	5.40 (137.2)
78	7.8 (198.1)	8.30 (210.8)	8.40 (213.4)	7.30 (185.4)	5.90 (149.9)
83	8.3 (210.8)	8.80 (223.5)	8.90 (226.1)	7.80 (198.1)	6.40 (162.6)
88	8.8 (223.5)	9.30 (236.2)	9.40 (238.8)	8.30 (210.8)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

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Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions (when applicable). Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Options. Card holders are available with the following options. Card holders not requiring any of the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 4). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

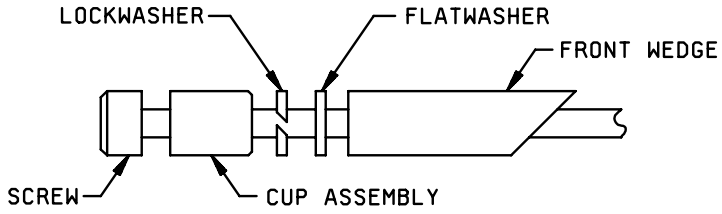


FIGURE 4. Lockwasher and flat washer details.

Screw self-locking element (see figure 5). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

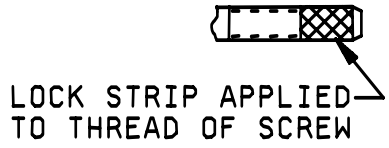


FIGURE 5. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of preparation and coordination of this CID specification sheet, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	APW Electronic Solutions 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: sales.electronicsolutions@apw.com Uniform Resource Locator (URL): www.apw.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. This CID specification sheet part numbers supersedes the following manufacturer's part numbers as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data. 1/

PIN designator AA59590/06	Vendor similar designator or type part number 2/ CAGE 18915	PIN designator AA59590/06	Vendor similar designator or type part number 2/ CAGE 52094
CJ28	40VI–6–LN	CJ28	V225CR2.80HR
CN28	40VI–6–LN–A	CN28	V225CR2.80R
CT28	40VI–6–LN–T	CT28	V225CR2.80T0R
CV28	40VI–6–LN–S	CV28	V225CR2.80T2R
CG28	40VI–6–LN–M	CG28	V225CR2.80TM2.5R
CJ28E	40VI–6–LN–L	CJ28E	V225CR2.80HLR
CN28E	40VI–6–LN–A–L	CN28E	V225CR2.80LR
CT28E	40VI–6–LN–T–L	CT28E	V225CR2.80T0LR
CV28E	40VI–6–LN–S–L	CV28E	V225CR2.80T2LR
CG28E	40VI–6–LN–M–L	CG28E	V225CR2.80TM2.5LR
CJ28W	40VI–6–LF–LN	CJ28W	WV225CR2.80HR
CN28W	40VI–6–LF–LN–A	CN28W	WV225CR2.80R
CT28W	40VI–6–LF–LN–T	CT28W	WV225CR2.80T0R
CV28W	40VI–6–LF–LN–S	CV28W	WV225CR2.80T2R
CG28W	40VI–6–LF–LN–M	CG28W	WV225CR2.80TM2.5R
CJ28D	40VI–6–LF–LN–L	CJ28D	WV225CR2.80HLR
CN28D	40VI–6–LF–LN–AL	CN28D	WV225CR2.80LR
CT28D	40VI–6–LF–LN–TL	CT28D	WV225CR2.80T0LR
CV28D	40VI–6–LF–LN–SL	CV28D	WV225CR2.80T2LR
CG28D	40VI–6–LF–LN–ML	CG28D	WV225CR2.80TM2.5LR

See footnotes at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/06	Vendor similar designator or type part number 2/ CAGE 18915	PIN designator AA59590/06	Vendor similar designator or type part number 2/ CAGE 52094
BJ28	40VI–6–B–LN	BJ28	VA225CR2.80HR
BN28	40VI–6–B–LN–A	BN28	VA225CR2.80R
BT28	40VI–6–B–LN–T	BT28	VA225CR2.80T0R
BV28	40VI–6–B–LN–S	BV28	VA225CR2.80T2R
BG28	40VI–6–B–LN–M	BG28	VA225CR2.80TM2.5R
BJ28E	40VI–6–B–LN–L	BJ28E	VA225CR2.80HLR
BN28E	40VI–6–B–LN–A–L	BN28E	VA225CR2.80LR
BT28E	40VI–6–B–LN–T–L	BT28E	VA225CR2.80T0LR
BV28E	40VI–6–B–LN–S–L	BV28E	VA225CR2.80T2LR
BG28E	40VI–6–B–LN–M–L	BG28E	VA225CR2.80TM2.5LR
BJ28W	40VI–6–B–LF–LN	BJ28W	WVA225CR2.80HR
BN28W	40VI–6–B–LF–LNA	BN28W	WVA225CR2.80R
BT28W	40VI–6–B–LF–LNT	BT28W	WVA225CR2.80T0R
BV28W	40VI–6–B–LF–LNS	BV28W	WVA225CR2.80T2R
BG28W	40VI–6–B–LF–LNM	BG28W	WVA225CR2.80TM2.5R
BJ28D	40VI–6–B–LFLNL	BJ28D	WVA225CR2.80HLR
BN28D	40VI–6–B–LFLNA	BN28D	WVA225CR2.80LR
BT28D	40VI–6–B–LFLNTL	BT28D	WVA225CR2.80T0LR
BV28D	40VI–6–B–LFLNSL	BV28D	WVA225CR2.80T2LR
BG28D	40VI–6–B–LFLNML	BG28D	WVA225CR2.80TM2.5LR
HJ28	40VI–6–B3–LN	HJ28	VHA225CR2.80HR
HN28	40VI–6–B3–LN–A	HN28	VHA225CR2.80R
HT28	40VI–6–B3–LN–T	HT28	VHA225CR2.80T0R
HV28	40VI–6–B3–LN–S	HV28	VHA225CR2.80T2R
HG28	40VI–6–B3–LN–M	HG28	VHA225CR2.80TM2.5R
HJ28E	40VI–6–B3–LN–L	HT28E	VHA225CR2.80HLR
HN28E	40VI–6–B3–LN–AL	HR28E	VHA225CR2.80LR
HT28E	40VI–6–B3–LN–TL	HS28E	VHA225CR2.80T0LR
HV28E	40VI–6–B3–LN–SL	HL28E	VHA225CR2.80T2LR
HG28E	40VI–6–B3–LN–ML	HG28E	VHA225CR2.80TM2.5LR
HJ28W	40VI–6–B3–LN–LF	HT28W	WVHA225CR2.80HR
HN28W	40VI–6–B3LF–LNA	HR28W	WVHA225CR2.80R
HT28W	40VI–6–B3LF–LNT	HS28W	WVHA225CR2.80T0R
HV28W	40VI–6–B3LF–LNS	HL28W	WVHA225CR2.80T2R
HG28W	40VI–6–B3LF–LNM	HG28W	WVHA225CR2.80TM2.5R

See footnotes at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/06	Vendor similar designator or type part number 2/ CAGE 18915	PIN designator AA59590/06	Vendor similar designator or type part number 2/ CAGE 52094
HJ28D	40VI–6–B3–LFLNL	HT28D	WVHA225CR2.80HLR
HN28D	40VI–6–B3LFLNAL	HR28D	WVHA225CR2.80LR
HT28D	40VI–6–B3LFLNTL	HS28D	WVHA225CR2.80T0LR
HV28D	40VI–6–B3LFLNSL	HL28D	WVHA225CR2.80T2LR
HG28D	40VI–6–B3LFLNML	HG28D	WVHA225CR2.80TM2.5LR

1/ CID PINs listed only with length designator “28”. See Table III for CID PIN construction using other available lengths.
2/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Example of PIN with available length designators.

PIN designator AA59590/06	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/06	Vendor similar designator or type part number 1/ 2/ CAGE 52094
CJ18	40VI–4–LN	CJ18	V225CR1.80HR
CJ23	40VI–5–LN	CJ23	V225CR2.30HR
CJ28	40VI–6–LN	CJ28	V225CR2.80HR
CJ30	40–5011–3.00	CJ30	V225CR3.00HR
CJ33	40VI–7–LN	CJ33	V225CR3.30HR
CJ35	40–5011–3.50	CJ35	V225CR3.50HR
CJ38	40VI–8–LN	CJ38	V225CR3.80HR
CJ43	40VI–9–LN	CJ43	V225CR4.30HR
CJ48	40VI–10–LN	CJ48	V225CR4.80HR
CJ53	40VI–11–LN	CJ53	V225CR5.30HR
CJ55	40–5011–5.50	CJ55	V225CR5.50HR
CK58	40VI–12–LN	CJ58	V225CR5.80HR
CK61	40–5011–6.10	CK61	V225CR6.10EHR
CK63	40VI–13–LN	CK63	V225CR6.30EHR
CK67	40–5011–6.70	CK67	V225CR6.70EHR
CK68	40VI–14–LN	CK68	V225CR6.80EHR
CK73	40VI–15–LN	CK73	V225CR7.30EHR
CK78	40VI–16–LN	CK78	V225CR7.80EHR
CK83	40VI–17–LN	CK83	V225CR8.30EHR
CK88	40VI–18–LN	CK88	V225CR8.80EHR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.

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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity
DLA – CC

Project 5998–0193-01

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